Question Paper ID: 678148

Course Code: CS354TA

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RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

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Mysuru Road Bengaluru - 560 059

V Semester B.E. Regular / Supplementary Examination February/ March 2026

COMMON TO CD / CS / CY / IS

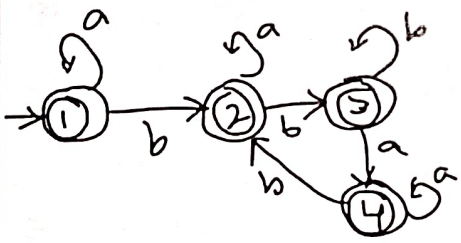
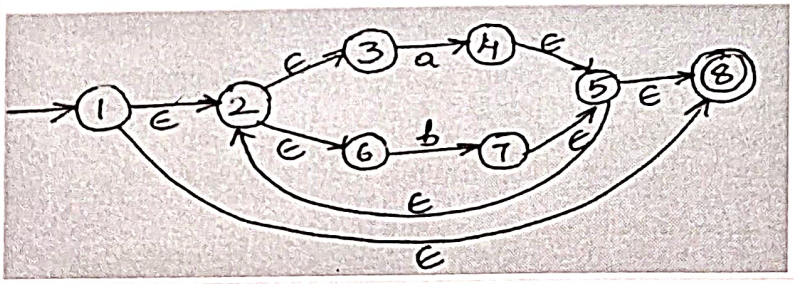
Theory of Computation

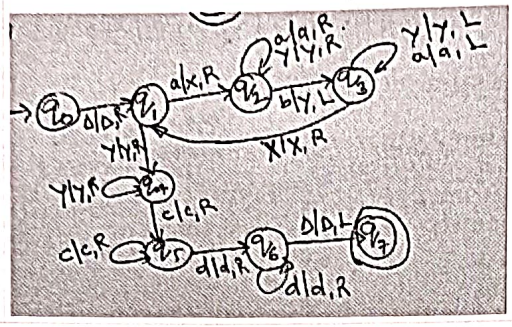
Time : 3 Hours **Maximum Marks : 100**

Instructions to the students

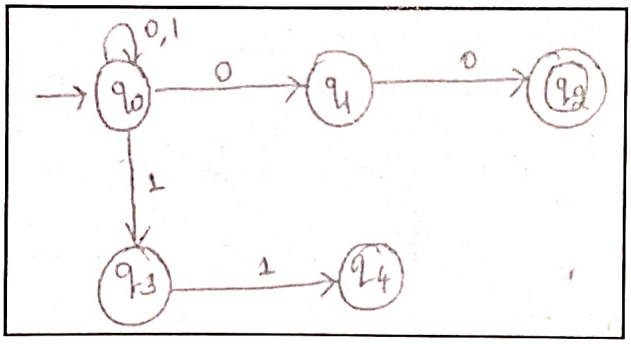
1. Answer all questions from Part A . Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Part A

Question No	Question	M	CO	BT
1.1	Left and right annihilator for concatenation is _____.	01	1	1
1.2	Minimize the following DFA. The number of states in minimized DFA is _____. 	02	4	3
1.3	Find the Ɛ-CLOSURE(1) 	01	3	1
1.4	Give an example of a string w for a language L = { 0^n1^n | n ≥ 1 } that could be pumped.	02	4	3
1.5	Eliminate left recursion from the following grammar E →E + T | T ,T →T * F | F , F → (E) | id	02	2	2
1.6	Decide whether the language L={w |w ϵ {a, b}* and na(w)=nb(w) } is a CFL or not? Justify your answer.	02	1	2
1.7	Define the language accepted by PDA.	02	1	1

1.8	Write the TM transition diagram for the given regular expression (0+1)*001.	02	2	2
1.9	Identify the last ID for (q_0ΔaabbcccddΔ) in the following Turing Machine. 	02	2	2
1.10	Prove that every context sensitive language is recursively enumerable language.	02	2	2
1.11	Define PCP.	02	2	2

Part B

Question No	Question	M	CO	BT
2a	Define NFA, Extended transition function of NFA and language accepted by NFA. For the NFA shown below, using the subset construction draw the equivalent DFA. 	08	4	3
2b	Define regular expression. Give regular expressions which generates the following languages over the alphabet Σ = {a,b}. i) The language of all strings in which the number of a's is even. ii)The language of all strings in which every a is followed immediately by bb. iii)The language of all strings in which both the number of a's and the number of b's are odd.	04	3	3
2c	For the regular expressions given below, find an Є – NFA recognizing language corresponds to the regular expressions. i) 0+10*+01*0 ii) (0+1)*(1+00)(0+1)*	04	3	1

3a	Define GNF.Convert the below grammar into GNF S→ AB, A → aA | bB | b , B → b	04	3	2
3b	Define regular Grammar .Give the automata for the following grammar S→0S|1A|1 , A→0A|1A|0|1	06	2	3
3c	Define CFG.Design a CFG for the language L(G)={ ab(bbaa)^n bba (ba)^n: n ≥0}	06	3	4
	OR			
4a	Obtain a CFG to generate the language L={ w | $n_a(w)=n_b(w)$}	04	2	3
4b	State and prove the Pumping Lemma for Regular Languages. Using it, prove that the language L = { $a^n b^n$ | n is odd } is not regular.	06	4	3
4c	Construct a finite automaton that accepts the language generated by the left linear grammar S→Ba | Ab | b , B →Ca | Bb , C→Aa | Cb | a	06	3	3
5a	Using CYK algorithm test whether string 110100 belongs to L(G) where G is given by: S → AB | BC A → BB | 0 B → BA | 1 C → AC | AA | 0	06	3	3
5b	Prove that language generated by empty stack PDA and final state PDA are equivalent.	06	2	2
5c	Prove that context free languages are not closed under intersection and complementation.	04	2	2
	OR			
6a	Define DPDA. Give DPDA to accept the following languages i) Language corresponds to (abb+b)*(ab)*. ii) L= {w | w ϵ {a, b}* and Na (w) ≠ Nb (w)}.	08	3	4
6b	Write the step by step procedure to find an equivalent CFG for the given PDA. Let M be the PDA whose transitions are listed below. Find the equivalent CFG for M. Show that the string w=babcbab is derivable. δ(q,a,Z)=(q,AZ), δ(q,b,Z)=(q,BZ), δ(q,a,A)=(q,AA), δ(q,b,A)=(q,BA), δ(q,a,B)=(q,AB), δ(q,b,B)=(q,BB), δ(q,c,Z)=(p,Z), δ(q,c,A)=(p,A), δ(q,c,B)=(p,B), δ(p,a,A)=(p,ϵ), δ(p,b,B)=(p,ϵ), δ(p,ϵ,Z)=(p,ϵ).	08	4	4
7a	If L_1 and L_2 are recursive languages over Σ, then show that $L_1 \cup L_2$ is also recursive.	04	2	2
7b	Explain with a neat sketch the working of restricted Turing Machine and Turing Machine with stay option.	06	1	1

7c	Construct Turing Machine for L = { $a^n b^m c^m$: n, m > 0 }. Trace the operation of TM for the string aaabbcc.	06	4	4
	OR			
8a	Design Turing Machine to perform the string reverse operation over Σ= {a, b, c}.	06	4	4
8b	If L_1 and L_2 are recursively enumerable languages over Σ, then show that L_1 - L_2 is also recursively enumerable languages.	04	2	2
8c	Explain the general structure of multitape & multidimensional TMs, show that those are equivalent to standard TM.	06	2	1
9a	If L_1 and L_2 are context sensitive languages over Σ, then show that $L_1 \cap L_2$ is also context sensitive languages.	04	2	2
9b	Define Context Sensitive Grammar (CSG). Write CSG to generate the language L = { ww^r : w $\in$ {a, b} *}. Give the derivation for the string aabbaa.	06	4	4
9c	Construct LBA for L = { w : $n_a(w) \neq n_b(w)$, w $\in$ {a, b} *}. Trace the operation of LBA for the string aabbabb.	06	4	4
	OR			
10a	Construct LBA to accepting the language. L={xxx | x$\in$ {a, b}*}. Using Instantaneous Descriptions show that the string ababab is accepted by LBA.	08	4	4
10b	Define Unrestricted Grammar (URG). Write URG to generate the language L={$a^i b^j c^k d^l$ | i=j and k=l $\geq$0}. Give the derivation for the string aabbcd.	08	4	4